Project Documentation Guide

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Building Evidence for Your Robot Project Report

> Many students in previous years submitted reports without adequate evidence of their work. This significantly limited the marks they could achieve. This guide shows you what documentation you need and how to collect it.

Why Documentation Matters

Engineers record procedures, observations and results systematically. **Your robot project requires the same approach.**

Instructors can only award marks for work that is demonstrated through evidence. Without documentation, even excellent work cannot be credited.

> Your robot is built as a team, but your report is individual. Keep your own copy of every photo, log and video, since you each write your own report from the same shared evidence.

What to Document

Essential Documentation Checklist

Approach and Reasoning	Sketches or diagrams, hand drawn or in AutoCAD if you prefer, showing decisions such as sensor placement, wiring layout or how you approached calibration, within the constraints of the kit

Building Progress	Photos of your robot at different stages, not just finished, showing the assembly sequence
Iteration and Testing Log	A record of each significant attempt, what you tried, what happened including failures, and what you changed as a result and why. This applies to both your code and your physical build.
Code Development	Multiple saved versions showing progression, for example v1, v2, v3, with commented code
Individual Contribution	Photos showing you working on specific tasks, with your own contribution clearly identified. Your report must include a short section describing your personal role, worth 10% of the report mark, separate from the team's work as a whole.

Keeping Your Iteration and Testing Log

Use the same approach you use in physics labs: a table to record trials, and diagrams to show your setup.

Testing Table Format

Create a table with one row per trial. Record observations as you test.

Trial	Test Conditions	Observations	Result	Notes
1	Both motors 50% speed, code v1	Robot veers left after 1 second, clicking from left motor	Failed	Motors may have different speeds
2	Swapped motors, same code	Problem stayed on left side, clicking moved to right motor	Failed	Confirms the motor itself has a problem

Trial	Test Conditions	Observations	Result	Notes
3	Replaced faulty motor, 50% speed	Straight line movement for a full 5 seconds, no clicking	Success	Motor replacement solved the issue

Evidence to support the record above: video clips from trials 1, 2 and 3 showing the problem and resolution, and a photo of the replaced motor.

Wiring and Setup Diagrams

Sketch your robot setup to show how components connect, motors, sensors and power, the wiring layout and pin connections, and the physical arrangement of parts.

These diagrams do not need to be perfect drawings. Clear, labelled sketches showing connections and layout are sufficient, whether hand drawn or created in AutoCAD. Take photos of your sketches, or create simple diagrams digitally if you prefer.

> Tables and diagrams are standard engineering documentation. They allow you to track systematic testing and clearly communicate your technical setup.

Using Evidence in Your Report

Reference your documentation explicitly. Compare these examples.

Poor, no evidence

"The first design did not work so we changed it."

Good, with evidence

"Our first prototype, Figure 3, failed because the motor mount was too flexible, causing 8mm deflection, Table 1. We added a triangular support bracket, Figure 4, reducing deflection to under 1mm."

Poor, no evidence

"We placed the ultrasonic sensor at the front of the robot."

Good, with evidence

"We mounted the ultrasonic sensor at the front centre of the chassis, Figure 2, after early testing,

Table 1, Trial 2, showed that an offset position produced inconsistent wall detection on turns."

Practical Documentation Tips

Photograph as you work

Capture work in progress, not just finished results. Date your photos.

Save all versions

Code: motor_control_v1.py, v2.py, v3.py. Sketches or diagrams: base_plate_sketch_v1.jpg, base_plate_sketch_v2.jpg, or base_plate_v1.dwg if using AutoCAD.

Document failures

Photos and notes about what did not work are valuable evidence of problem solving.

Make notes immediately

Do not rely on memory. Write down decisions and changes as you make them.

Keep it organised

Use a shared team folder with a clear structure, and keep your own copy for your report.

Document regularly

Make documentation a habit every session, not something you do at the end.

Engineering is not about getting it right the first time.

It is about systematic problem solving and learning from failures. Your documentation proves you did this properly.